
Cross-City Generalization in Airbnb Price Prediction in Top Tourism Cities within the U.S.

Zoe Zhou¹ Lucia Jiang¹

Abstract

We evaluate cross-city generalization in Airbnb price prediction across five major US cities. Despite shared socioeconomic patterns, local market dynamics often degrade model performance in unseen locations. We test tree-based ensembles and neural models under a Leave-One-City-Out protocol, assessing accuracy, error range, and stability. While single-city transfers exhibit a significant generalization gap, calibrated grouped transfer substantially reduces variance, enabling reliable cross-city estimation.

1. Introduction

Airbnb is an important part of the U.S. short-term rental market, especially in tourism cities. It provides flexible accommodation options for travelers, but Airbnb listings differ widely in room type, capacity, host behavior, and location. Compared with hotels, Airbnb listings are less standardized, which makes it harder to define a single reference market price. Therefore, price prediction depends on both listing-level features and local market conditions.

At the same time, cities are not completely independent markets. Urban science research has shown that socioeconomic indicators often follow systematic scaling patterns across cities (Bettencourt et al., 2007). While Airbnb price prediction is often treated as a standard supervised learning problem in which the goal is to predict the nightly price, in practice, a model may perform well within the trained cities, but fail when applied to another where local housing conditions, tourism demand, and listing composition are very different. These cities share common listing structures, but also different in market condition, urban density, and tourist geography, making the transfer performance study crucial.

In this study, we evaluate cross-city generalization in

Airbnb price prediction across five major U.S. tourism cities: Boston, Los Angeles (LA), New York City (NYC), San Francisco (SF), and Seattle. First, we train within-city baseline and tree-based models to establish city-specific benchmarks. We then test single-city transfer, where a model trained on one city is applied to another city. To better understand transfer failure modes, we also evaluate calibrated transfer and grouped-city training, including leave-one-city-out (LOCO) models. Finally, we compare tree-based models with a multilayer perceptron (MLP) and use permutation feature importance to examine whether the same predictors remain stable and influential across within-city prediction and cross-city transfer.

Our results show that cross-city generalization is possible but uneven. Tree-based models improve substantially over linear baselines within cities, but single-city transfer often produces large performance gaps. Grouped and calibrated transfer can improve stability for some target cities, but they do not fully eliminate city-specific errors. Feature importance analysis further suggests that basic structural features, such as room type and accommodates, are more transferable, while host, review, and availability-related features are less stable across cities.

2. Literature Review

Prior work has shown that Airbnb prices are influenced by multiple types of listing-level and market-level factors. Wang & Nicolau (2017) studied Airbnb listings from 33 cities worldwide and used ordinary least squares (OLS) and quantile regression to examine price determinants, including host attributes, property characteristics, rental rules, and review ratings. Their findings show that Airbnb pricing is explained by a combination of determinants rather than single dominant factor, and reveal heterogeneous pricing patterns across different price levels and country-level geographic heterogeneity. Related work on hotel pricing also shows that local spatial competition affects accommodation prices. Balaguer & Pernías (2013) find that higher local competitor density is associated with lower average hotel prices and lower price dispersion in urban hotels. This supports our use of a tourist area indicator that captures spatial market differences in high-demand areas.

¹Center for Data Science, New York University, New York, United States. Correspondence to: Xiaoli (Zoe) Zhou <xz3514@nyu.edu>, Qike (Lucia) Jiang <qj2169@nyu.edu>.

Recent studies have also applied machine learning (ML) methods to Airbnb price prediction. Pittala et al. (2024) analyzed Airbnb prices across ten major European cities using linear regression, polynomial regression, and Random Forest. Their results show that Random Forest outperforms the regression-based models, which suggests nonlinear relationship between listing features and price. Moreover, this study highlights the importance of geographic factors, which motivated our use of tourist area indicator. Likewise, Liu (2021) compared multiple statistical and ML models for Airbnb price prediction, and found that Gradient Boosting and XGBoost models outperform other models. In particular, XGBoost achieved the best performance, which supports our use of ensemble tree-based models as the main nonlinear models.

However, these studies mainly focus on explaining price determinants or improving prediction accuracy, but do not directly evaluate whether models trained on one city or a group of cities can generalize to unseen cities. Our project focuses on filling this gap by evaluating the cross-city generalization across five major U.S. tourism cities, with particular attention to analyzing transfer failure modes and whether important predictors remain stable across within-city prediction and cross-city transfer.

3. Data

3.1. Data Source

We retrieved listing data from Inside Airbnb (Inside Airbnb, 2025), where all 5 cities’ data are from the September 2025 snapshot to ensure temporal consistency. After dropping samples with missing price values, our dataset contains a total of 73,482 listings (Boston:3506; LA: 36819; NYC: 21141; SF: 5795; Seattle: 6221), with 80 features including listings’ identification, property information, geographic information, host information, reviews, city, and price.

3.2. Data Preprocessing

To prepare the data for modeling, we excluded `Hotel room` and `Shared room` from the modeling dataset due to the limited sample sizes, shown in Figure 1, which makes it difficult for models to learn reliable patterns. In addition, these two categories are conceptually different from the two main Airbnb listing types, `Entire home/apt` and `Private room`.

We transformed `host_since` into `host_years`, representing host tenure as of September 1, 2025.

Furthermore, To evaluate cross-city generalization, we introduced a boolean variable, `tourist_area`, to represent location desirability. For each city we identified a set of tourist neighborhoods based on TripAdvisor, and matched

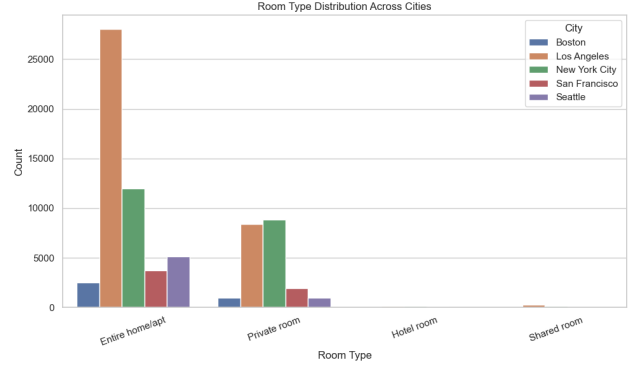


Figure 1. Room type distribution across cities before filtering. Hotel room and shared room accounts for a very small share of the data.

them to the actual neighborhood labels in the dataset. Seattle has 7 neighborhoods identified as tourist neighborhoods due to geological limitation, while the other four cities each have 10.

We imputed missing values in `host_response_rate`, `host_acceptance_rate`, and `host_years` with city-wise mean. Missing values in the boolean variable `host_has_profile_pic` were filled with 0, effectively designating the absence of a profile picture as the default. For listings missing `bathrooms` value, we first extracted the numeric bathroom count from `bathrooms_text`. After this, we dropped the 24 listings whose `bathrooms` values remained missing. Feature `accommodates` has no missing values, thus we used it to impute missing values in `beds` and `bedrooms` based on ration of `beds-to-accommodates` and `bedrooms-to-accommodates` from the ones with complete `accommodates`, `beds`, and `bedrooms` values.

Finally, our target variable `price` is highly right-skewed, as shown in Figure 2 where `log_price` is on the x-axis. To reduce the influence of extreme price outliers, we removed listings with prices above the city-specific 99th percentile.

After all preprocessing, the final modeling dataset consists 71,732 listings with no missing values (Boston:3453; LA: 35817; NYC: 20675; SF: 5650; Seattle: 6137). There are 14 predictors: property information `room_type`, `accommodates`, `bathrooms`, `bedrooms`, and `beds`; booking features `minimum_nights` and `availability_365`; review features `number_of_reviews` and `review_scores_rating`; host information `host_response_rate`, `host_acceptance_rate`, `host_has_profile_pic` and `host_years`; geographic boolean indicator `tourist_area`. In addition,



Figure 2. Log price distribution across cities before applying 99th percentile price filter.

`city` is retained for cross-city training and evaluation, while `price` and `log_price` are the original and transformed target variables for modeling.

4. Methods

4.1. Baselines

To provide reference performance, we trained simple baseline models within each city. All baseline models were trained to predict log-transformed price, consistent with the tree-based and multilayer perceptron (MLP) neural network models trained later on. We applied one-hot encoding to the only categorical feature `room_type`, and applied standard scaling to all 13 numerical features. We trained five baseline models, each of which was evaluated within each city using 5-fold cross-validation.

The first two are DummyRegressor models, which predict the target using only the training fold’s central tendency, including a mean baseline and a median baseline. These models do not use any other listing-level features as they are intended to represent a lower-bound reference for model performance.

We also included linear regression, Ridge regression, and Lasso regression. These models use the same set of final features as the tree-based and MLP models, but assume a mostly linear relationship between listing characteristics and price. Adding Ridge and Lasso regularization controls overfitting and provides a stronger comparison than using ordinary linear regression alone.

These baselines help answer two questions: whether listing features are useful for price prediction, and whether more flexible nonlinear models provide better prediction performance.

4.2. Single-city Modeling

We further created within-city benchmark by training models separately within each city using Random Forest and XGBoost, both of which used the same final feature set as baselines. We trained models on three target variants, `log_price`, `mean_norm_log_price`, and `median_norm_log_price`, where the latter two examine whether city-level price scale differences affect predictions.

For each city, tree-based models were evaluated using an outer 5-fold cross-validation. Hyperparameters were selected using randomized search inside the training fold, which allowed us to tune model parameters only on the training portion of the specific fold.

4.3. Calibrated-transfer Modeling

We next evaluated a calibrated transfer setting to test whether cross-city transfer errors are mainly driven by city-level price scale differences or the feature-price relationship. Tree-based models were trained on the full source city data and applied to each of the other cities as target cities.

While using the same feature set, we shifted each target-city numeric feature by the difference between the source-city feature mean and target-city feature mean before applying the source-city scaler to the target-city data. Predictions were then re-anchored using the target city’s mean or median price, helps separate price-level mismatch from differences in feature-price relationships across cities.

4.4. Grouped-city Modeling

Furthermore, we tested whether training on multiple cities improves transfers, using Random Forest and XGBoost. The grouped-city models were trained using the same final feature set and the same three target variants. For grouped tree-based transfers, numerical features were calibrated following the same procedure described in the previous section.

We trained grouped-city models and tested on held-out cities, with two types of groups considered. First, we tested selected empirical groups based on observed transfer patterns, which are explained in the results section. These groups include LA with Seattle, Boston with SF, and LA with Seattle and SF. Secondly, we used a leave-one-city-out (LOCO) design, where the model was trained on four cities and tested on the remaining one.

4.5. MLP Modeling

We also trained a feed-forward multilayer perceptron (MLP) model as a higher-capacity comparison model to test if a more flexible neural network can improve cross-city generalization beyond the tree-based models.

The MLP was evaluated mainly in grouped-city settings rather than conducting all single-city transfer pairs. This choice is made because of the limited data size we have for Boston, SF, and Seattle, which could potentially cause model instability if we train them separately since neural networks are more sensitive to limited training data. Grouped-city training provides a larger and more diverse training set, which is more suitable for testing the MLP.

The MLP used the same final feature set and target variants as the other models. Hyperparameters including learning rate and dropout were selected from a small candidate set. These experiments are used as a comparison to the tree-based transfer models.

4.6. Feature Importance Analysis

Finally, to better interpret model behaviors, we computed permutation importance for selected models. Permutation importance measures how much prediction error increases when a feature is randomly shuffled. Features that cause larger increases in error are treated as more important for prediction.

We computed permutation importance using MAE in terms of log-price. The analysis compares within-city XGBoost models, LOCO XGBoost models, and LOCO MLP models. Features that remain important across settings are interpreted as more general Airbnb pricing factors, while features whose importance changes across settings are interpreted as more city-specific. This allows us to examine whether the same features remain important in both city-specific prediction and cross-city transfer.

5. Results

For presentation clarity and data integrity, additional descriptive details are deferred to the Appendix. Throughout this section, we denote Los Angeles as LA, New York City as NYC, and San Francisco as SF. Moreover, RF denotes Random Forest, XGB denotes XGBoost, LP denotes log price, MN denotes mean-normalized log price, and MD denotes median-normalized log price.

5.1. Within-city Prediction Performance

Table 1. Best within-city performance for baseline linear models

City	Best Baseline	MAE
Boston	Linear Reg.	80.3
LA	Linear Reg.	96.0
NYC	Linear Reg.	97.6
SF	Ridge	96.2
Seattle	Ridge	54.6

Tables 1 and 2 summarize the best-performing baseline

Table 2. Best within-city performance for tree-based models

City	Best Tree	Target	MAE	RMSE
Boston	Random Forest	Mean-norm	56.2	106.8
LA	XGBoost	Mean-norm	74.5	148.2
NYC	Random Forest	Mean-norm	65.0	130.5
SF	Random Forest	Log price	64.3	125.4
Seattle	XGBoost	Log price	48.1	96.3

linear model and the best-performing tree-based model for each city, selected by MAE.

Tree-based models achieve lower MAE than the best linear baseline in all five cities, indicating that nonlinear models capture important pricing patterns beyond linear feature. These improvements suggest that nonlinear effects and feature interactions, such as relationships among location, property size, room type, and review signals, play an important role in Airbnb pricing across these five cities.

Among all tree-based models, Random Forest performs best in Boston, NYC, and SF, while XGBoost performs best in LA and Seattle. The best target variant differs slightly across cities, but the resulting MAE values are similar across target transformations, suggesting the target normalization has limited impact in the within-city setting.

5.2. Single-city Cross-city Transfer

Table 3. Single-city cross-city transfer MAE for tree-based models

Train	Boston	LA	NYC	SF	Seattle
Boston	—	108.3 _{RF-LP}	98.1 _{XGB-MN}	88.7 _{XGB-LP}	88.8 _{XGB-MD}
LA	102.3 _{XGB-LP}	—	110.2 _{RF-MD}	92.3 _{XGB-LP}	63.1 _{RF-MN}
NYC	88.1 _{XGB-MN}	127.6 _{RF-MD}	—	95.6 _{XGB-MN}	132.0 _{RF-MD}
SF	77.1 _{RF-LP}	106.9 _{RF-MD}	98.5 _{RF-LP}	—	93.1 _{RF-MD}
Seattle	111.9 _{XGB-MD}	90.5 _{XGB-MN}	112.2 _{RF-MN}	106.4 _{XGB-MD}	—

Table 4. Single-city cross-city transfer within-50% accuracy for tree-based models

Train	Boston	LA	NYC	SF	Seattle
Boston	—	0.549 _{RF-LP}	0.692 _{XGB-MN}	0.720 _{XGB-LP}	0.539 _{XGB-MD}
LA	0.752 _{XGB-LP}	—	0.693 _{RF-MD}	0.794 _{XGB-LP}	0.764 _{RF-MN}
NYC	0.701 _{XGB-MN}	0.466 _{RF-MD}	—	0.635 _{XGB-MN}	0.337 _{RF-MD}
SF	0.832 _{RF-LP}	0.581 _{RF-MD}	0.698 _{RF-LP}	—	0.554 _{RF-MD}
Seattle	0.688 _{XGB-MD}	0.742 _{XGB-MN}	0.647 _{RF-MN}	0.721 _{XGB-MD}	—

The above table 3 and 4 summarize the best-performance in MAE and within-50% accuracy in out-of-sample test among models Random Forests and XGBoost along with 3 different target variants. This experiment measures whether pricing patterns learned from one city generalize to another. Information and data about the Spearman Rank can be found in table 7 in the appendix.

The cross-city transfer results show a clear decrease in performance compared with within-city predictions. For exam-

ple, Seattle has a within-city tree MAE of 48.1, but models trained on Boston, NYC, or SF produce much larger MAE values when transferred to Seattle. Similarly, Boston has a within-city tree MAE of 56.2, while models trained on LA or Seattle produce MAE values above 100 when tested on Boston. This suggests that although tree-based models capture useful within-city pricing patterns, models trained on a single source city often learn market-specific relationships that do not fully transfer to other markets, and performance varies a lot by different train-test city pairs.

The within-50% accuracy results provide a more practical view of this transfer gap. Some city pairs achieve relatively strong transfer performance: for example, SF to Boston reaches a within-50% accuracy of 0.832 and LA to Seattle reaches 0.764. However, other transfer pairs perform much worse, such as NYC to Seattle with a within-50% accuracy of only 0.337 and NYC to LA with 0.466. These results indicate that cross-city transfer is not uniformly difficult; rather, transfer success depends strongly on the relationship between the source and target city markets.

Importantly, the transfer patterns are also asymmetric. Training on LA and testing on Seattle gives a relatively low MAE of 63.1, while training on Seattle and testing on LA gives a higher MAE of 90.5. Similarly, training on SF transfers relatively well to Boston, with MAE of 77.1 and within-50% accuracy of 0.832, while training on Boston and testing on SF gives a higher MAE of 88.7 and lower within-50% accuracy of 0.720. Therefore, geographic domain shift depends on both the pricing structure learned from the source city and the feature distribution of the target city.

5.3. Grouped-city Cross-city Transfer with Calibration

5.3.1. TREE-MODEL-BASED

Based on the single-city transfer results, we further grouped cities by transfer similarity, defined by how well one city generalizes to another using metrics such as MAE, RMSE, and within-25% accuracy. We then evaluated whether training on multiple source cities improves transfer robustness. In addition to empirically selected city groups, we also conducted leave-one-city-out (LOCO) grouped transfer, where each city is held out as the test city and the model is trained on the remaining four cities.

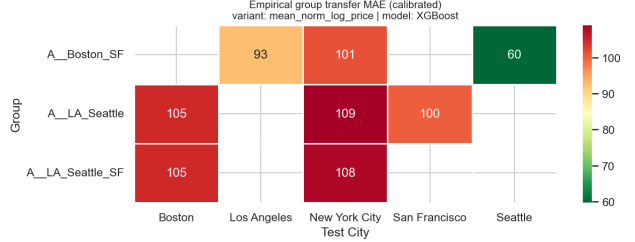


Figure 3. Empirical grouped-city transfer performance for selected source-city groups

Table 5. Calibrated LOCO grouped transfer performance for tree-based models

Test City	Target	MAE	Within-50%	Spearman
Boston	Median-norm	83.1	0.715	0.748
LA	Median-norm	93.5	0.707	0.776
NYC	Log price	102.8	0.646	0.670
SF	Median-norm	94.2	0.693	0.702
Seattle	Mean-norm	55.5	0.842	0.738

Figure 3 summarizes the performance of the empirically selected grouped-city models. It shows that some grouped source sets can produce reasonable transfer performance, but the benefit is not uniform across all target cities. This suggests that grouping cities can help only when the source cities contain pricing relationships that are relevant to the target market.

We can also find the calibrated LOCO grouped transfer results for tree-based models in table 5. Among the held-out cities, Seattle has the strongest LOCO grouped performance, with MAE of 55.5 and within-50% accuracy of 0.842. Boston also improves compared with many single-city transfer results, with LOCO grouped MAE of 83.1. In contrast, New York City remains the most difficult target city, with LOCO grouped MAE of 102.8 and within-50% accuracy of 0.646. This indicates that even after calibration and multi-city training, some city markets remain harder to generalize to.

Figure 7, shown in appendix, compares the within-city benchmark, the best single-city transfer result, and the LOCO grouped result for each target city. Grouped training is not uniformly better than the best single-city transfer. It improves or remains competitive for some targets, such as Boston and Seattle, but performs worse for New York City and San Francisco.

Table 6. MLP LOCO grouped transfer performance

Test City	Target	MAE	Within-50%	Spearman
Boston	Mean-norm	90.2	0.786	0.748
LA	Median-norm	100.8	0.630	0.758
NYC	Mean-norm	101.2	0.713	0.744
SF	Median-norm	93.2	0.754	0.711
Seattle	Mean-norm	69.4	0.699	0.694

5.3.2. MLP-BASED

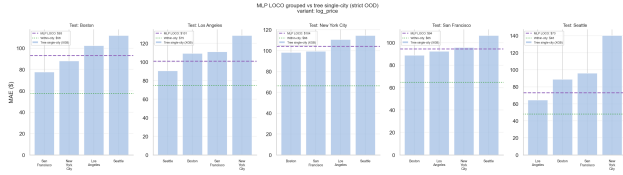


Figure 4. MLP LOCO grouped transfer compared with tree-based single-city transfer and within-city benchmarks

We also evaluated MLP models under the LOCO grouped transfer setting to test whether a more flexible neural network model improves cross-city generalization. Table 6 shows the best MLP result for each held-out city, selected by MAE across target variants. The MLP achieves its lowest transfer error for Seattle, with MAE of 69.4, and an highest error for LA, with MAE of 100.8. The within-50% results show a similar patterns: Boston and SF have relatively strong practical accuracy, while LA and Seattle are less stable under MLP transfer.

Figure 4 compares MLP LOCO grouped transfer with the best tree-based single-city transfer and the within-city benchmark. Overall, the MLP does not consistently outperform the tree-based transfer models, and also requiring greater computational cost. Other metrics for MLP could also be found in table 8 in the appendix.

5.4. Feature Importance

To better understand why some models transfer better than the others, we analyze permutation feature importance under both within-city prediction and LOCO cross-city transfer. Figure 5 compares XGBoost feature importance across these two settings. In the within-city setting, room type, accommodates, minimum nights, host years, and bathrooms are among the most important predictors. Under LOCO transfer, however, the importance of most features decreases substantially. This suggests that many predictors are useful within a local market but become less stable when transferred across cities, however the relative rank is more stable.

The most transferable features are still related to basic listing structure. Room type and accommodates remain the strongest predictors under LOCO transfer, indicating that

XGBoost feature importance under within-city vs cross-city transfer

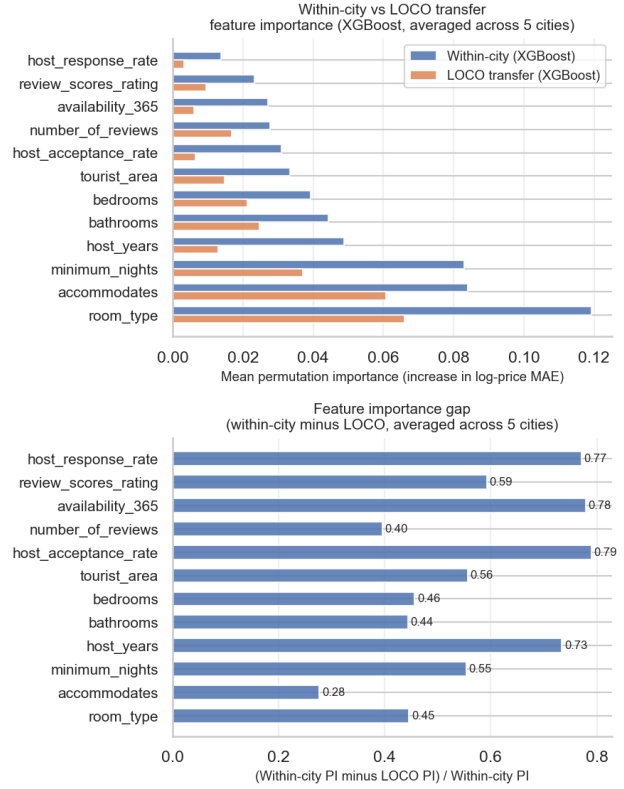


Figure 5. XGBoost permutation feature importance under within-city prediction and LOCO cross-city transfer. Feature importance is averaged across the five cities and measured as the increase in log-price MAE after permuting each feature.

broad property characteristics generalize more consistently across cities. In contrast, features such as availability, host acceptance rate, host response rate, and review scores show larger importance gaps between within-city and LOCO settings. These features may reflect more city-specific market behavior, platform usage patterns, or local demand conditions, making them less reliable for cross-city predictions.

Figure 6 compares LOCO feature importance between XGBoost and MLP. The comparison highlights two consistently transferable predictors: `room_type` and `accommodates`. The secondary signals are also clear, and both models agree on `minimum_nights` and `bathrooms`. However, several other features show lower permutation importance in the MLP, suggesting that the neural model relies more heavily on `accommodates` and `room_type` solely.

6. Evaluation

Our evaluations assess prediction accuracy and cross-city generalization to determine model transferability. More

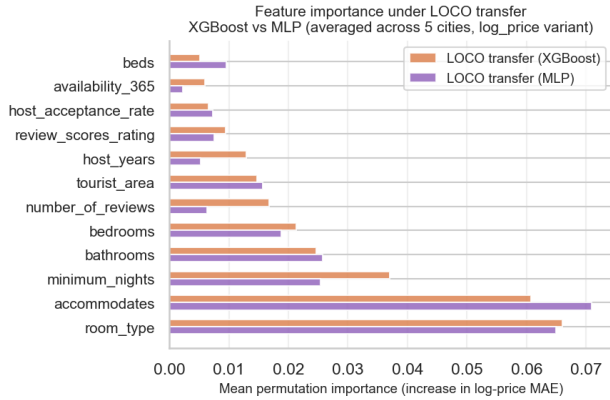


Figure 6. Permutation feature importance under LOCO transfer for XGBoost and MLP with averaged across the five cities using the log-price target

specifically, we analyze performance variance when training and testing configurations differ.

6.1. Interpretation of Performance Metrics

While models are trained on log-transformed targets, we back-transform all predictions to dollar values for evaluation. This ensures practical interpretability and consistent cross-city comparisons. We use MAE as the primary metric because it directly measures the average dollar-value error.

We also report RMSE to leverage its mechanism of penalizing large prediction errors more heavily. This is particularly important for Airbnb price prediction because the price distribution remains right-skewed even after initial outlier treatment. A higher RMSE compared to MAE indicates that while the model performs well for certain listings, it may struggle with expensive or unique "luxury" cases.

Median Absolute Error (MedAE) is included to provide a robust measure of typical error that is resistant to the influence of extreme values. Comparing MAE, RMSE, and MedAE together clarifies whether poor performance stems from a general error or a small subset of extreme cases.

Furthermore, we use Spearman rank correlation which assesses relative price ordering to capture systematic socioeconomic patterns, maintaining model utility during zero-shot transfer even as exact pricing shifts. Additionally, within-25% and within-50% accuracy quantify the proportion of predictions meeting practical utility thresholds.

6.2. Cross-City Generalization and Domain Shift

Strong within-city performance does not guarantee strong cross-city transfer. Tree-based models substantially improve over linear baselines within each city, which suggests that nonlinear relationships and feature interactions are important for Airbnb price prediction. However, when models

trained on one city are directly applied to another city, performance often drops. This indicates that Airbnb pricing patterns are not universal across cities.

This performance gap stems from the geographic domain shift. Variations in local price scales, housing supply, and tourism patterns alter the mapping between listing features and price. Consequently, identical attributes—such as `room_type` or `bedrooms`, corresponding to different valuation levels across cities like NYC, Seattle, and LA.

Single-city results reveal directional asymmetry. For example, models trained in Seattle generalize effectively to Boston, but the reverse transfer yields significant degradation. This suggests transferability depends on the source city’s representativeness. Diverse, heterogeneous markets capture pricing patterns applicable to smaller targets, whereas the limited variation in smaller source cities fails to generalize to complex urban environments.

Grouped-city and LOCO experiments evaluate whether multi-city training improves generalization. Grouped training can reduce transfer instability through exposure to diverse pricing patterns, though improvements are not uniform across all targets. This suggests that simply increasing training volume is insufficient; source cities must contain pricing relationships relevant to the target city. Thus, successful cross-city generalization depends on both data diversity and market similarity.

6.3. Model Interpretability

Permutation feature importance is used to evaluate which predictors contribute most to model performance and whether these predictors remain stable across modeling settings.

The feature importance results show that broad property characteristics, especially `room_type` and `accommodates`, are the most transferable predictors across cities. This finding is intuitive because these features reveals the reality of how people search and filter places. A full apartment generally differs from a private room, and listings that accommodate more guests tend to have higher prices regardless of city.

In contrast, some host, review, and availability-related features appear less stable across cross-city transfer settings. These variables may reflect local platform behavior, city-specific demand patterns, or differences in how hosts manage listings across markets. For example, availability or host response behavior may have different meanings in a year-long high-tourism city compared with a more seasonal or supply-constrained market, making these features only useful for within-city predictions.

Tree-based models offer clearer feature importance patterns

and achieve competitive or better transfer performance in many cases. The MLP model has greater flexibility, but it does not consistently outperform tree-based models and is also harder to interpret.

6.4. Robustness Checks

We conduct several robustness checks to evaluate whether the main findings are stable across modeling choices. First, models are trained using three target variants: log price, mean-normalized log price, and median-normalized log price. These variants are included because Airbnb price levels differ across geographic markets, and normalization allows us to examine whether cross-city transfer failure is mainly driven by city-level price scale differences. The results show that normalization helps in some cases, but it does not fully remove the transfer gap. This suggests that cross-city prediction error is caused not only by differences in overall price levels, but also by differences in feature-price relationships across cities.

Second, we compare Random Forest and XGBoost to test whether the results depend on a specific tree-based algorithm. The best-performing model differs across cities and experimental settings, but the broader pattern is consistent: tree-based models perform strongly in within-city prediction while still showing clear degradation under cross-city transfer. This supports the conclusion that the observed generalization gap is not caused by one particular tree model, but reflects a broader geographic domain shift in Airbnb pricing.

Third, grouped-city and leave-one-city-out experiments test whether using more source-city data improves transfer robustness. These experiments show that multi-city training can reduce transfer error for some target cities, but the benefits are uneven. In several cases, grouped training performs worse than the best single-city source model. This suggests that more training data is not automatically better; the similarity between the source and target city markets is often more important than the total amount of training data.

6.5. Limitations

There are several limitations to this study. First, the data is from a single time snapshot, so might not capture seasonal variations. Although we included the feature `availability_365` to represent each listing's availability in the next 365 days, it could not reveal all information, for example some people prefer to book until the last minute. Airbnb prices may vary across months, holidays, and local events, so future work could evaluate whether cross-city generalization changes over time.

Secondly, the current feature set focuses mainly on listing-level characteristics and a simplified tourist-area indicator.

While these features are useful and available across cities, they may not fully capture neighborhood-level economic conditions, transportation accessibility, local regulations, or tourism intensity. These omitted factors may explain part of the remaining cross-city generalization gap. At the same time, features like `beds` and `bedrooms` are related, but they do also show different perspectives, and thus how to mix and combine them is still an issue as well.

Third, the tourist-area variable is manually constructed based on selected neighborhoods. Although this provides a simple way to include geographic context, it may not capture finer neighborhood differences within each city. More detailed spatial features, such as distance to major attractions or neighborhood-level average price, could improve model performance but may also reduce transferability if they are too city-specific.

Fourth, the grouped-city experiments are based on five U.S. tourism cities. This provides a useful comparison across different urban markets, but the findings may not generalize to smaller cities, international cities, or non-tourism-heavy locations. A larger set of cities would allow a more systematic analysis of which market characteristics make transfer successful.

Finally, although calibration and grouped training improve robustness in some cases, they do not fully eliminate domain shift. This means that cross-city Airbnb price prediction should be used carefully in practice. A transferred model may provide useful approximate rankings or broad price estimates, but reliable dollar-level prediction still requires local market information.

7. Conclusion

This project evaluates Airbnb price prediction as a cross-city generalization problem across popular U.S. tourism cities. Tree-based models outperform linear baselines within cities, but generally lose prediction accuracy when tested on unseen cities. Employing a more complex MLP model with the same features does not necessarily improve the transfer performance. Grouped-city training with calibration gives mixed results. However, feature importance analysis shows that basic structural predictors, such as `room_type` and `accommodates` remain generally stable.

Our results suggest that Airbnb pricing contains transferable structure, but also strong city-specific patterns. Overall, cross-city Airbnb price prediction is possible but unstable, local market differences play an important role in pricing behavior.

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A. Appendix

Table 7. Single-city cross-city transfer Spearman rank correlation for tree-based models.

Train	Boston	LA	NYC	SF	Seattle
Boston	—	0.759 _{RF-LP}	0.713 _{XGB-MN}	0.741 _{XGB-LP}	0.715 _{XGB-MD}
LA	0.756 _{XGB-LP}	—	0.711 _{RF-MD}	0.767 _{XGB-LP}	0.710 _{RF-MN}
NYC	0.740 _{XGB-MN}	0.758 _{RF-MD}	—	0.726 _{XGB-MN}	0.693 _{RF-MD}
SF	0.808 _{RF-LP}	0.788 _{RF-MD}	0.738 _{RF-LP}	—	0.743 _{RF-MD}
Seattle	0.724 _{XGB-MD}	0.775 _{XGB-MN}	0.666 _{RF-MN}	0.699 _{XGB-MD}	—

Table 8. Full MLP LOCO grouped transfer metrics across target variants.

Test City	Target	MAE	RMSE	Med. AE	R^2	Spearman	Within-25%	Within-50%	Overprice
Boston	Log price	93.1	154.6	49.1	0.322	0.744	0.427	0.780	0.682
Boston	Mean-norm	90.2	149.8	49.7	0.364	0.748	0.431	0.786	0.658
Boston	Median-norm	91.6	151.7	50.1	0.347	0.758	0.418	0.794	0.664
LA	Log price	100.9	183.2	57.1	0.430	0.746	0.349	0.617	0.435
LA	Mean-norm	101.0	190.5	55.2	0.383	0.753	0.353	0.623	0.440
LA	Median-norm	100.8	265.8	54.3	-0.202	0.758	0.357	0.630	0.457
NYC	Log price	104.1	192.9	43.1	0.164	0.728	0.364	0.705	0.762
NYC	Mean-norm	101.2	187.6	42.1	0.209	0.744	0.372	0.713	0.741
NYC	Median-norm	102.2	188.7	43.9	0.200	0.743	0.374	0.712	0.753
SF	Log price	94.3	170.9	46.4	0.332	0.689	0.426	0.745	0.610
SF	Mean-norm	93.7	169.4	44.7	0.344	0.701	0.429	0.757	0.594
SF	Median-norm	93.2	170.5	43.5	0.336	0.711	0.434	0.754	0.583
Seattle	Log price	73.1	117.6	49.0	0.263	0.718	0.375	0.649	0.563
Seattle	Mean-norm	69.4	115.2	45.6	0.294	0.694	0.395	0.699	0.509
Seattle	Median-norm	71.2	117.4	45.4	0.266	0.703	0.395	0.686	0.530

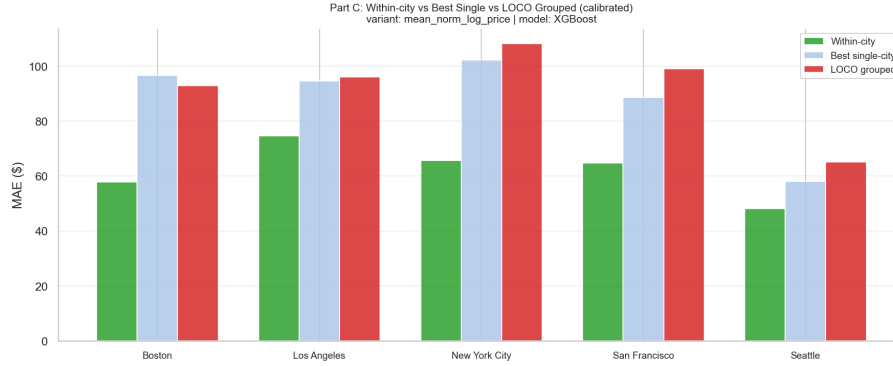


Figure 7. Comparison of within-city, best single-city transfer, and calibrated LOCO grouped transfer MAE by target city